

Exploratory Data Analysis

```
library(haven)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
dat25 <- read_xpt("../data/LLCP2025.XPT")
dat24 <- read_xpt("../data/LLCP2024.XPT")
#dat23 <- read_xpt("../data/LLCP2023.XPT")
dat22 <- read_xpt("../data/LLCP2022.XPT")
#dat21 <- read_xpt("../data/LLCP2021.XPT")
#dat20 <- read_xpt("../data/LLCP2020.XPT")
dat19 <- read_xpt("../data/LLCP2019.XPT")
#dat18 <- read_xpt("../data/LLCP2018.XPT")
dat17 <- read_xpt("../data/LLCP2017.XPT")
```

Exploratory data analysis

variables asked only in 2025, 2024, 2022, 2019, and 2017

NOBCUSE9 = What was main reason for not doing anything to keep you from getting pregnant?

PFPPRVN4 = The last time you had sexual intercourse, did you or your partner do anything

to keep you from getting pregnant? () **FAMILY PLANNING** = The last time you had sexual intercourse, what did you or your partner do to keep you from getting pregnant? (lots of response categories!!!)

variables asked in all surveys

SEXVAR = 2 if female

MARITAL= marital status (1=married, 2=divorced, 3=widowed, 4=separated, 5=never married, 6=A member of an unmarried couple, 9 =refused)

HOME= homeowner status (1=own, 2=rent, 3=other arrangement, 7=don't know, 9=refused)

PREGNANT = current pregnancy status (1=yes, 2=no, 7=don't know, 9=refused)

MENTHLTH = Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? (1-30, 88=none, 77=don't know, 99=refused)

EMPLOY = employment status (1 = employed for wages, 2 = self-employed, 3 = out of work for 1 year or more, 4 = out of work for less than 1 year, 5 = homemaker, 6 = student, 7 = retired, 8 = unable to work, 9 = refused)

INCOME = income categories (1=less than 15k, 2=15k to 25k, 3= 25k to 35k, 4= 35k to 50k, 5=50k to 100k, 6=100k to 200k, 7=200k or more, 9=don't know)

EDUCA = What is the highest grade or year of school you completed? (1=no high school, 2=high school, 3=some college, 4=college, 9=don't know)

INSURANCE = Respondents aged 18-64 who have any form of health insurance (1=yes, 2=no, 9=don't know)

RACE = Race/ethnicity categories (1=nh white, 2=nh black, 3=nh native american, 4=nh asian, 5=nh native hawaiian, 7=nh multiracial, 9=don't know)

subset each survey year to include relevant variables, subsetting the sample to only women

Recode the states in each year since states that didn't collect enough data for a given year weren't included in the dataset

```

sub_dat25 <- dat25 |> select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH",
  ↳ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE", "RENTHOM1",
  ↳ "MARITAL", "TYPCNT10")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↳ rename(INCOME='_INCOMG1', INSURANCE='_HCVU654', EMPLOY='EMPLOY1',
  ↳ FAMILY_PLANNING="TYPCNT10", "HOME"=RENTHOM1, RACE="_RACE")

state_labels_2025 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",
  `41` = "Oregon",
  `42` = "Pennsylvania",

```

```

`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)

sub_dat25 <- sub_dat25 |>
  dplyr::mutate(
    `STATE` = as.character(unname(
      state_labels_2025[as.character(`STATE`)]
    ))
  ) |>
  mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

sub_dat24 <- dat24 |> select(c("STATE", "SEXVAR", "PREGNANT", "MENTHLTH",
  ↪ "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU654", "_RACE", "RENTHOM1", "MARITAL",
  ↪ "TYPCNTR9")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG1', INSURANCE='_HCVU654', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR9", "HOME"=RENTHOM1, RACE="_RACE")

state_labels_2024 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",

```

```
`8` = "Colorado",
`9` = "Connecticut",
`10` = "Delaware",
`11` = "District of Columbia",
`12` = "Florida",
`13` = "Georgia",
`15` = "Hawaii",
`16` = "Idaho",
`17` = "Illinois",
`18` = "Indiana",
`19` = "Iowa",
`20` = "Kansas",
`21` = "Kentucky",
`22` = "Louisiana",
`23` = "Maine",
`24` = "Maryland",
`25` = "Massachusetts",
`26` = "Michigan",
`27` = "Minnesota",
`28` = "Mississippi",
`29` = "Missouri",
`30` = "Montana",
`31` = "Nebraska",
`32` = "Nevada",
`33` = "New Hampshire",
`34` = "New Jersey",
`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
```

```

`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico",
`78` = "Virgin Islands"
)
sub_dat24 <- sub_dat24 |>
  dplyr::mutate(
    `STATE` = as.character(unname(
      state_labels_2024[as.character(`STATE`)]
    ))
  ) |> mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

# Roe v. Wade was overturned on June 24, 2022
sub_dat22 <- dat22 |>
  ↪ select(c(`STATE`, "SEXVAR", "IMONTH", "IDAY", "IYEAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE1",
  ↪ "TYPCNTR9", "RENTHOM1")) |> filter(SEXVAR==2) |> select(-SEXVAR) |>
  ↪ rename(INCOME='_INCOMG1', INSURANCE='_HCVU652', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR9", HOME="RENTHOM1", RACE="_RACE1")
state_labels_2022 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",

```

```

`19` = "Iowa",
`20` = "Kansas",
`21` = "Kentucky",
`22` = "Louisiana",
`23` = "Maine",
`24` = "Maryland",
`25` = "Massachusetts",
`26` = "Michigan",
`27` = "Minnesota",
`28` = "Mississippi",
`29` = "Missouri",
`30` = "Montana",
`31` = "Nebraska",
`32` = "Nevada",
`33` = "New Hampshire",
`34` = "New Jersey",
`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico",
`78` = "Virgin Islands"
)
sub_dat22 <- sub_dat22 |>
  dplyr::mutate(
    `STATE` = as.character(unname(

```

```

state_labels_2022[as.character(`_STATE`)]
)
))|> mutate(EDUCA = case_when(
  EDUCA %in% 1:3 ~ 1, # no high school
  EDUCA == 4 ~ 2, # high school
  EDUCA == 5 ~ 3 ,#Some college
  EDUCA == 6 ~ 4 ,#College graduate
  EDUCA == 9 ~ 9 #Don't know
))

```

```

sub_dat19 <- dat19 |>
  ↪ select(c("_STATE", "SEXVAR", "PREGNANT", "MENTHLTH", "EMPLOY1", "_INCOMG",
  ↪ "EDUCA", "_HCVU651", "_RACE", "RENTHOM1", "MARITAL", "TYPCNTR8")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
  ↪ INSURANCE='_HCVU651', EMPLOY='EMPLOY1', FAMILY_PLANNING="TYPCNTR8",
  ↪ "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2019 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",

```

```

`29` = "Missouri",
`30` = "Montana",
`31` = "Nebraska",
`32` = "Nevada",
`33` = "New Hampshire",
`35` = "New Mexico",
`36` = "New York",
`37` = "North Carolina",
`38` = "North Dakota",
`39` = "Ohio",
`40` = "Oklahoma",
`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)
sub_dat19 <- sub_dat19 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2019[as.character(`_STATE`)]
    ))
  )
sub_dat19 |>
  mutate(EDUCA = case_when(
    EDUCA %in% 1:3 ~ 1, # no high school
    EDUCA == 4 ~ 2, # high school
    EDUCA == 5 ~ 3, #Some college
    EDUCA == 6 ~ 4, #College graduate
    EDUCA == 9 ~ 9 #Don't know
  ))

```

```

sub_dat17 <- dat17 |> select(c("_STATE", "SEX", "PREGNANT", "MENTHLTH",
  ↪ "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
  ↪ "MARITAL", "TYPCNTR7")) |> filter(SEX==2) |> select(-SEX) |>
  ↪ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1',
  ↪ FAMILY_PLANNING="TYPCNTR7", "HOME"=RENTHOM1, RACE="_RACE")
state_labels_2017 <- c(
  `1` = "Alabama",
  `2` = "Alaska",
  `4` = "Arizona",
  `5` = "Arkansas",
  `6` = "California",
  `8` = "Colorado",
  `9` = "Connecticut",
  `10` = "Delaware",
  `11` = "District of Columbia",
  `12` = "Florida",
  `13` = "Georgia",
  `15` = "Hawaii",
  `16` = "Idaho",
  `17` = "Illinois",
  `18` = "Indiana",
  `19` = "Iowa",
  `20` = "Kansas",
  `21` = "Kentucky",
  `22` = "Louisiana",
  `23` = "Maine",
  `24` = "Maryland",
  `25` = "Massachusetts",
  `26` = "Michigan",
  `27` = "Minnesota",
  `28` = "Mississippi",
  `29` = "Missouri",
  `30` = "Montana",
  `31` = "Nebraska",
  `32` = "Nevada",
  `33` = "New Hampshire",
  `34` = "New Jersey",
  `35` = "New Mexico",
  `36` = "New York",
  `37` = "North Carolina",
  `38` = "North Dakota",
  `39` = "Ohio",
  `40` = "Oklahoma",

```

```

`41` = "Oregon",
`42` = "Pennsylvania",
`44` = "Rhode Island",
`45` = "South Carolina",
`46` = "South Dakota",
`47` = "Tennessee",
`48` = "Texas",
`49` = "Utah",
`50` = "Vermont",
`51` = "Virginia",
`53` = "Washington",
`54` = "West Virginia",
`55` = "Wisconsin",
`56` = "Wyoming",
`66` = "Guam",
`72` = "Puerto Rico"
)
sub_dat17 <- sub_dat17 |>
  dplyr::mutate(
    `_STATE` = as.character(unname(
      state_labels_2017[as.character(`_STATE`)]
    ))
  )
mutate(EDUCA = case_when(
  EDUCA %in% 1:3 ~ 1, # no high school
  EDUCA == 4 ~ 2, # high school
  EDUCA == 5 ~ 3, #Some college
  EDUCA == 6 ~ 4, #College graduate
  EDUCA == 9 ~ 9 #Don't know
))

```

```

sub_dat23 <- dat23 |> select(c("_STATE", "SEXVAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU653", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU653', EMPLOY='EMPLOY1')

sub_dat21 <- dat21 |> select(c("_STATE", "SEXVAR", "PREGNANT",
  ↪ "MENTHLTH", "EMPLOY1", "_INCOMG1", "EDUCA", "_HCVU652", "_RACE")) |>
  ↪ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG1',
  ↪ INSURANCE='_HCVU652', EMPLOY='EMPLOY1')

```

```

sub_dat20 <- dat20 |> select(c("_STATE", "SEXVAR", "PREGNANT",
  ↳ "MENTHLTH", "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE")) |>
  ↳ filter(SEXVAR==2) |> select(-SEXVAR) |> rename(INCOME='_INCOMG',
  ↳ INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

sub_dat18 <- dat18 |> select(c("_STATE", "SEX1", "PREGNANT", "MENTHLTH",
  ↳ "EMPLOY1", "_INCOMG", "EDUCA", "_HCVU651", "_RACE", "RENTHOM1",
  ↳ "MARITAL")) |> filter(SEX1==2) |> select(-SEX1) |>
  ↳ rename(INCOME='_INCOMG', INSURANCE='_HCVU651', EMPLOY='EMPLOY1')

```

merge relevant questionnaire years

```

data_list <- list(
  `2017` = sub_dat17,
  `2019` = sub_dat19,
  `2022` = sub_dat22,
  `2024` = sub_dat24,
  `2025` = sub_dat25
)

vars_to_keep <- c(
  "_STATE",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",
  "EDUCA",
  "INSURANCE",
  "RACE",
  "MARITAL",
  "HOME",
  "FAMILY_PLANNING",
  "IYEAR",
  "IMONTH",
  "IDAY"
)

analysis_data <- purrr::imap_dfr(
  data_list,

```

```

function(dataset, year) {

  year <- as.integer(year)

  dataset <- dataset |>
    dplyr::select(dplyr::any_of(vars_to_keep)) |>
    dplyr::mutate(
      YEAR = year
    )

  if (year == 2022) {
    dataset <- dataset |>
      dplyr::mutate(
        IYEAR = as.integer(as.character(IYEAR)),
        IMONTH = as.integer(as.character(IMONTH)),
        IDAY = as.integer(as.character(IDAY)),

        date = as.Date(
          sprintf("%04d-%02d-%02d", IYEAR, IMONTH, IDAY)
        ),

        post_ban = dplyr::case_when(
          date < as.Date("2022-06-24") ~ 0L,
          date >= as.Date("2022-06-24") ~ 1L,
          TRUE ~ NA_integer_
        )
      )
  } else {
    dataset <- dataset |>
      dplyr::mutate(
        date = as.Date(NA),

        POST_BAN = dplyr::case_when(
          YEAR <= 2021 ~ 0L,
          YEAR >= 2023 ~ 1L,
          TRUE ~ NA_integer_
        )
      )
  }

  dataset
}
) |>

```

```
dplyr::rename(STATE = `_STATE`) |> select(-c(IYEAR, IMONTH, IDAY, date))

# check that post ban was correctly created
analysis_data |> filter(YEAR == 2025) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
NA	187696	1	NA

```
analysis_data |> filter(YEAR == 2024) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
NA	240183	1	NA

```
analysis_data |> filter(YEAR == 2022) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
0	96964	4.110508e-01	0.411056
1	138926	5.889365e-01	0.588944
NA	3	1.271763e-05	NA

```
analysis_data |> filter(YEAR == 2019) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
NA	228419	1	NA

```
analysis_data |> filter(YEAR == 2017) |> janitor::tabyl(post_ban)
```

post_ban	n	percent	valid_percent
NA	251007	1	NA

create U.S. census regions

```

library(tibble)

census_region_lookup <- tribble(
  ~STATE, ~REGION, ~DIVISION,

  # Northeast
  "Connecticut",      "Northeast", "New England",
  "Maine",            "Northeast", "New England",
  "Massachusetts",    "Northeast", "New England",
  "New Hampshire",    "Northeast", "New England",
  "Rhode Island",     "Northeast", "New England",
  "Vermont",          "Northeast", "New England",
  "New Jersey",       "Northeast", "Middle Atlantic",
  "New York",          "Northeast", "Middle Atlantic",
  "Pennsylvania",     "Northeast", "Middle Atlantic",

  # Midwest
  "Illinois",         "Midwest", "East North Central",
  "Indiana",          "Midwest", "East North Central",
  "Michigan",         "Midwest", "East North Central",
  "Ohio",             "Midwest", "East North Central",
  "Wisconsin",        "Midwest", "East North Central",
  "Iowa",             "Midwest", "West North Central",
  "Kansas",           "Midwest", "West North Central",
  "Minnesota",        "Midwest", "West North Central",
  "Missouri",         "Midwest", "West North Central",
  "Nebraska",         "Midwest", "West North Central",
  "North Dakota",     "Midwest", "West North Central",
  "South Dakota",     "Midwest", "West North Central",

  # South
  "Delaware",         "South", "South Atlantic",
  "District of Columbia", "South", "South Atlantic",
  "Florida",          "South", "South Atlantic",
  "Georgia",          "South", "South Atlantic",
  "Maryland",         "South", "South Atlantic",
  "North Carolina",   "South", "South Atlantic",
  "South Carolina",   "South", "South Atlantic",
  "Virginia",         "South", "South Atlantic",
  "West Virginia",    "South", "South Atlantic",
  "Alabama",          "South", "East South Central",
  "Kentucky",         "South", "East South Central",
  "Mississippi",      "South", "East South Central",

```

```

"Tennessee",      "South", "East South Central",
"Arkansas",       "South", "West South Central",
"Louisiana",      "South", "West South Central",
"Oklahoma",       "South", "West South Central",
"Texas",          "South", "West South Central",

# West
"Arizona",        "West", "Mountain",
"Colorado",       "West", "Mountain",
"Idaho",          "West", "Mountain",
"Montana",        "West", "Mountain",
"Nevada",         "West", "Mountain",
"New Mexico",     "West", "Mountain",
"Utah",           "West", "Mountain",
"Wyoming",        "West", "Mountain",
"Alaska",         "West", "Pacific",
"California",     "West", "Pacific",
"Hawaii",         "West", "Pacific",
"Oregon",         "West", "Pacific",
"Washington",     "West", "Pacific",

# Territories: not assigned to a Census region/division
"Guam",           NA_character_, NA_character_,
"Puerto Rico",    NA_character_, NA_character_,
"Virgin Islands", NA_character_, NA_character_
)

# remove Guam, Puerto Rico, and Virgin Islands from analytic sample
analysis_data_v2 <- analysis_data |>
  dplyr::left_join(
    census_region_lookup,
    by = "STATE"
  ) |> filter(!is.na(REGION))

```

```

variable_labels <- list(

  MARITAL = c(
    `1` = "Married",
    `2` = "Divorced",
    `3` = "Widowed",
    `4` = "Separated",
    `5` = "Never married",

```

```

`6` = "Member of an unmarried couple",
`9` = "Refused"
),

HOME = c(
  `1` = "Own",
  `2` = "Rent",
  `3` = "Other arrangement",
  `7` = "Don't know",
  `9` = "Refused"
),

PREGNANT = c(
  `1` = "Yes",
  `2` = "No",
  `7` = "Don't know",
  `9` = "Refused"
),

MENTHLTH = c(
  setNames(
    paste(1:30, ifelse(1:30 == 1, "day", "days")),
    as.character(1:30)
  ),
  `77` = "Don't know",
  `88` = "None",
  `99` = "Refused"
),

EMPLOY = c(
  `1` = "Employed for wages",
  `2` = "Self-employed",
  `3` = "Out of work for 1 year or more",
  `4` = "Out of work for less than 1 year",
  `5` = "Homemaker",
  `6` = "Student",
  `7` = "Retired",
  `8` = "Unable to work",
  `9` = "Refused"
),

INCOME = c(
  `1` = "Less than $15,000",

```

```

    `2` = "$15,000 to $24,999",
    `3` = "$25,000 to $34,999",
    `4` = "$35,000 to $49,999",
    `5` = "$50,000 to $99,999",
    `6` = "$100,000 to $199,999",
    `7` = "$200,000 or more",
    `9` = "Don't know"
  ),

  EDUCA = c(
    `1` = "No high school",
    `2` = "High school",
    `3` = "Some college",
    `4` = "College graduate",
    `9` = "Don't know"
  ),

  INSURANCE = c(
    `1` = "Yes",
    `2` = "No",
    `9` = "Don't know"
  ),

  RACE = c(
    `1` = "Non-Hispanic White",
    `2` = "Non-Hispanic Black",
    `3` = "Non-Hispanic Native American",
    `4` = "Non-Hispanic Asian",
    `5` = "Non-Hispanic Native Hawaiian",
    `6` = "Other race, non-Hispanic",
    `7` = "Non-Hispanic Multiracial",
    `8` = "Hispanic",
    `9` = "Don't know"
  )
)

variables_to_summarize <- c(
  "MARITAL",
  "HOME",
  "PREGNANT",
  "MENTHLTH",
  "EMPLOY",
  "INCOME",

```

```

"EDUCA",
"INSURANCE",
"RACE"
)

```

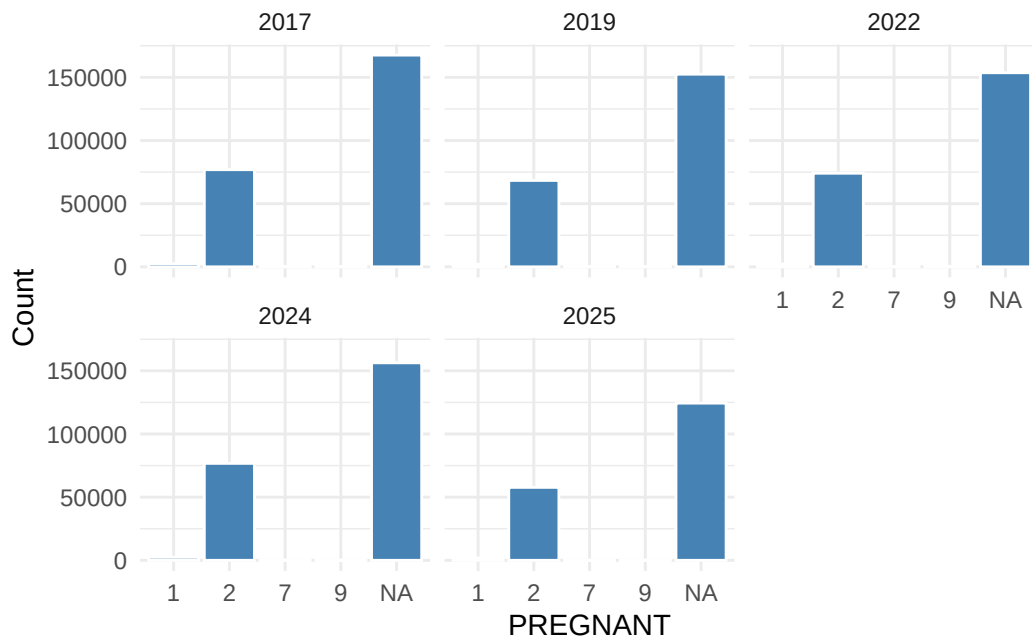
U.S. Population Characteristics Over Time

```

# categorical variables
sapply(names(analysis_data_v2)[!names(analysis_data_v2) %in%
↪ c("STATE","YEAR","POST_BAN", "MENTHLTH")], function(var){
  ggplot(
    analysis_data_v2,
    aes(x = factor(.data[[var]]))
  ) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    x = var,
    y = "Count"
  )
})

```

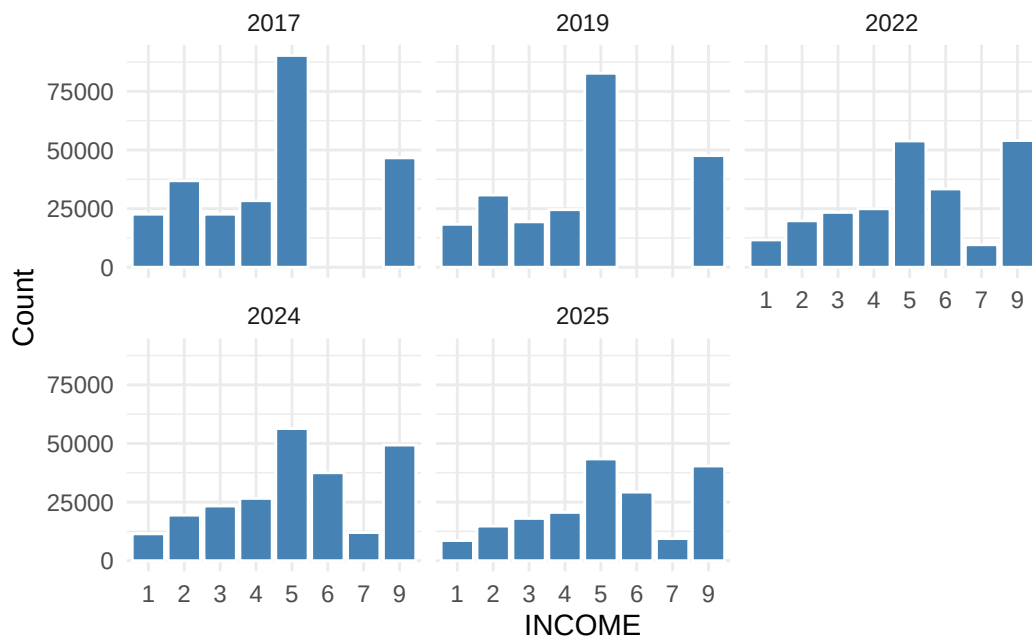
\$PREGNANT



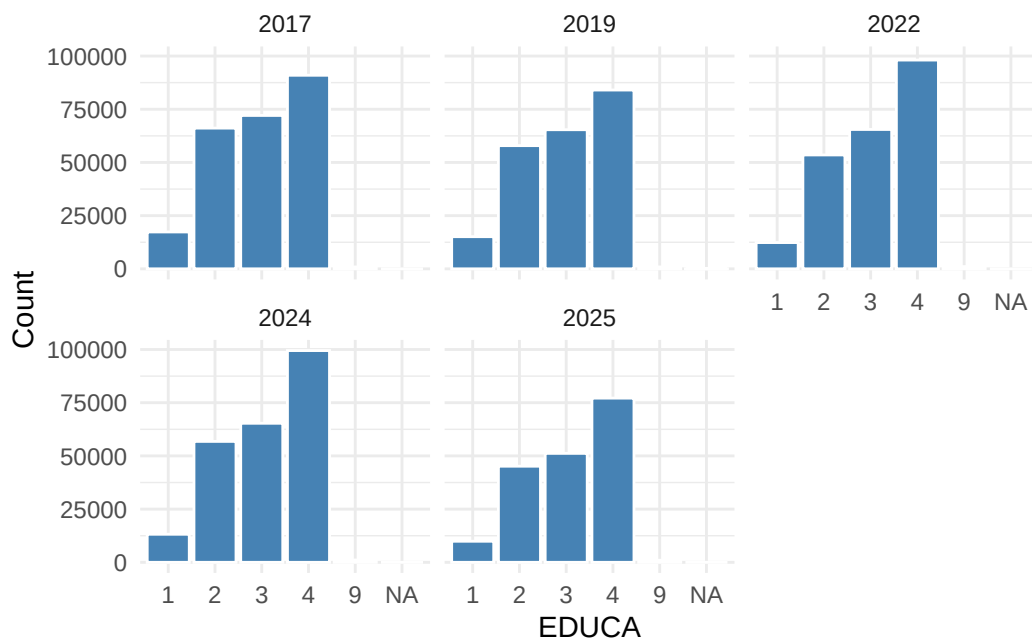
\$EMPLOY



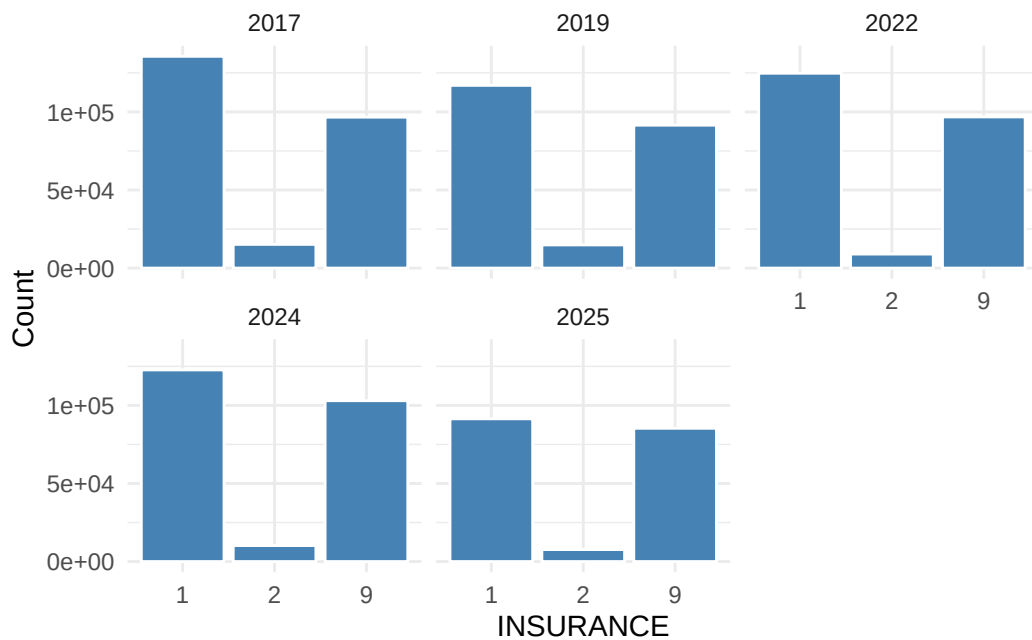
\$INCOME



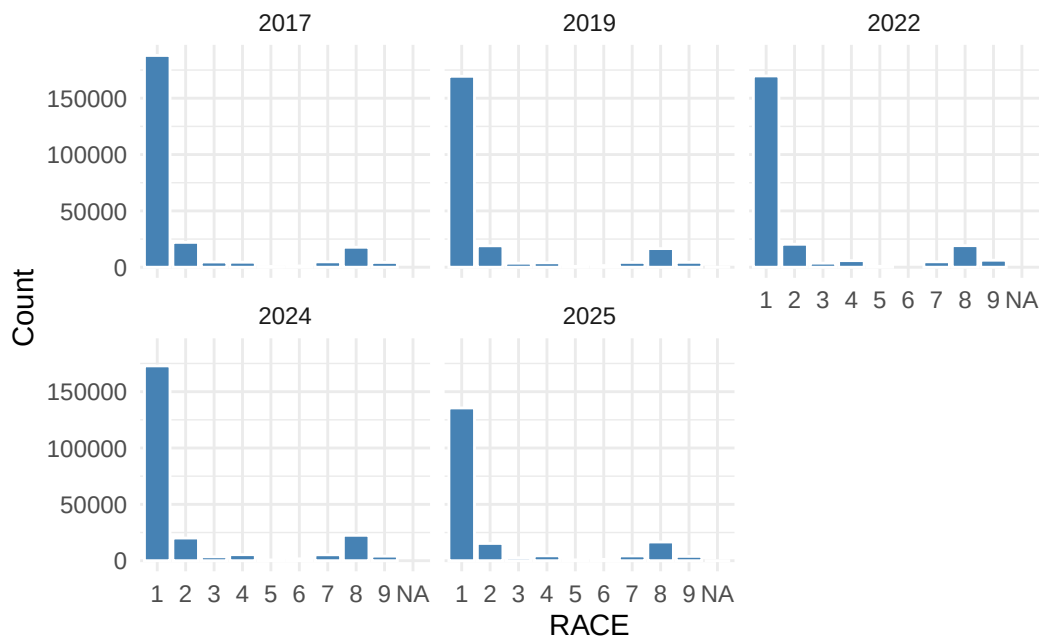
\$EDUCA



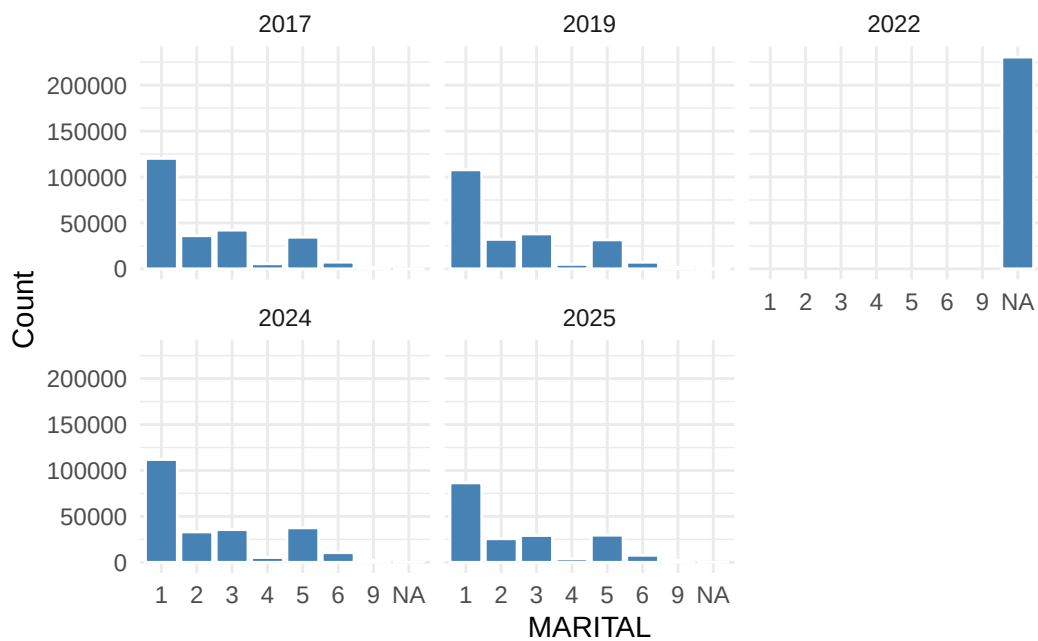
\$INSURANCE



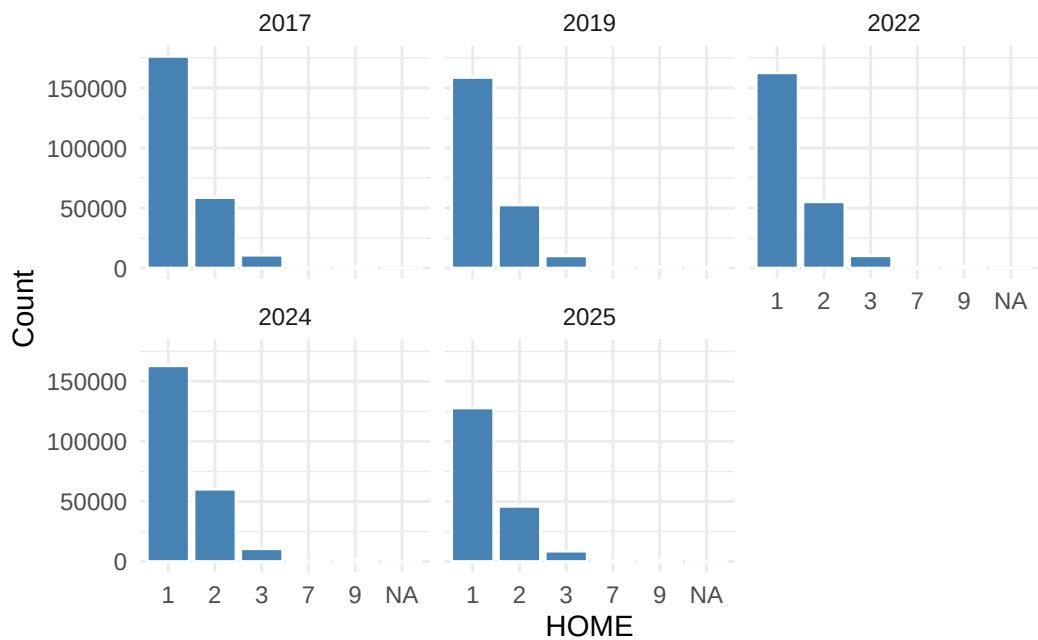
\$RACE



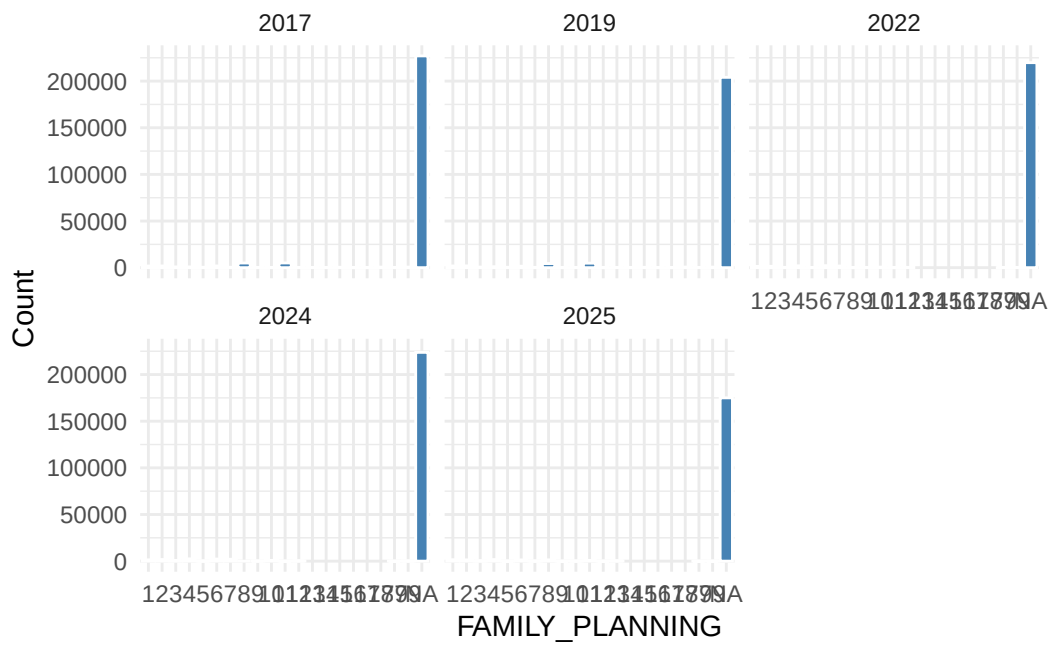
\$MARITAL



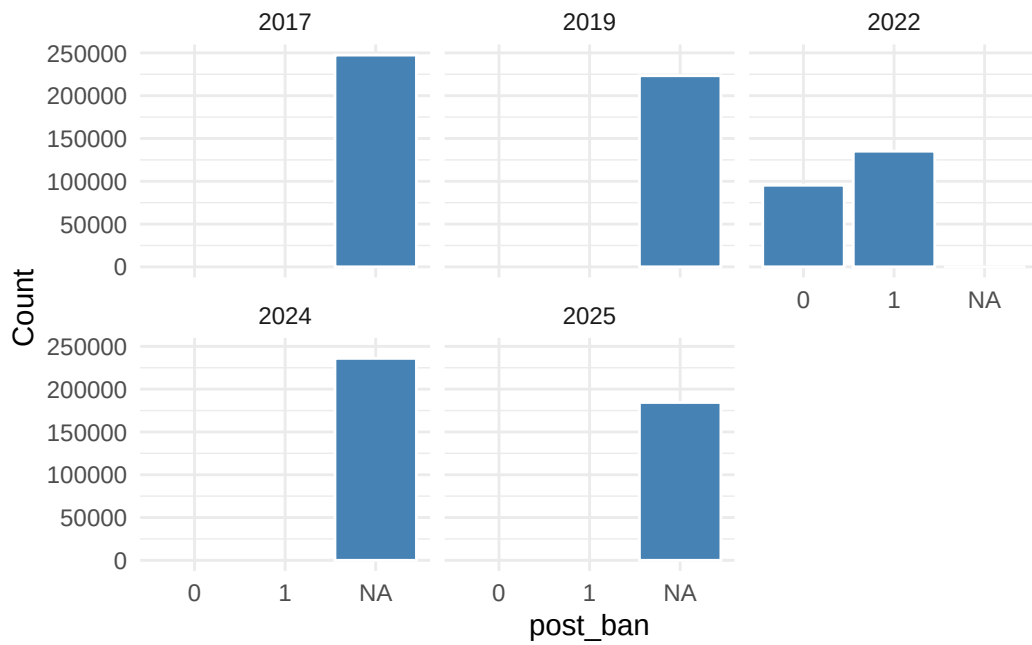
\$HOME



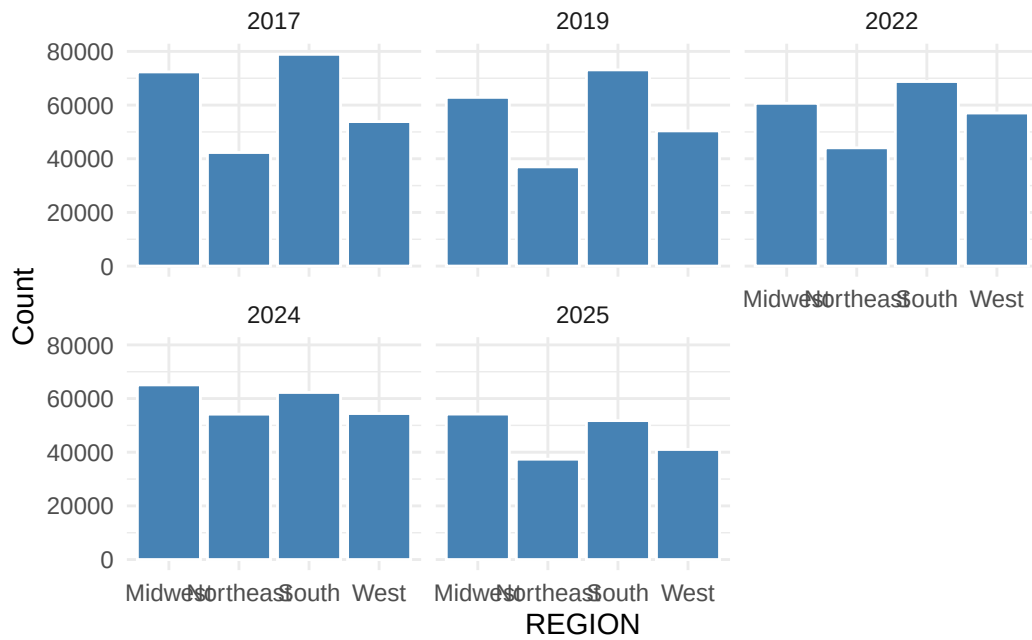
\$FAMILY_PLANNING



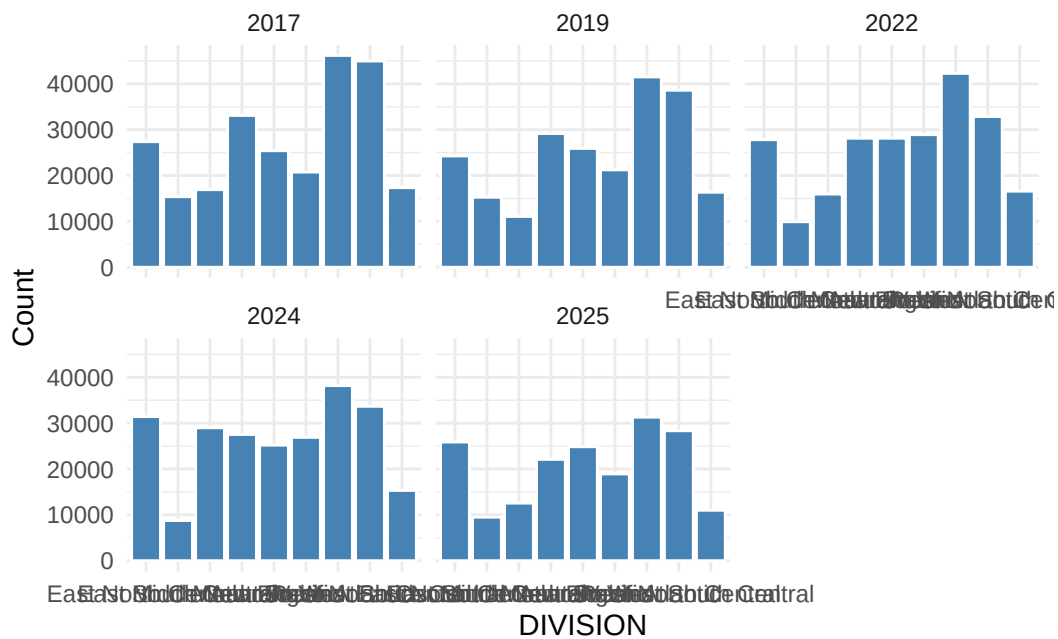
\$post_ban



\$REGION



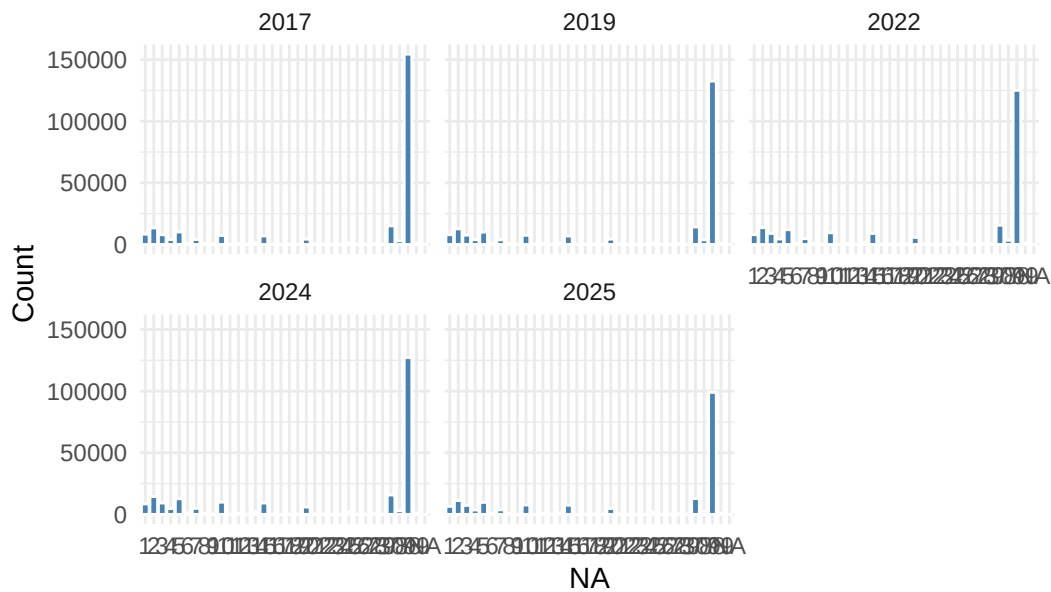
\$DIVISION



```
# numerical variable
ggplot(
  analysis_data_v2 ,
  aes(x = factor(MENTHLTH))
) +
  geom_bar(
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ YEAR) +
  theme_minimal() +
  labs(
    title = "Distribution of Days with Poor Mental Health",
    x = var,
    y = "Count"
  )
```

Warning in label(labels[[nm]] %||% ""): NAs introduced by coercion

Distribution of Days with Poor Mental Health



```
get_descriptive_stats_over_entire_period <- function(data, var) {

  tab <- data |>
    janitor::tabyl(
      {{ var }},
      show_na = TRUE
    )

  # The first column is the categorical variable
  category_col <- names(tab)[1]

  total_n <- sum(tab$n)

  valid_n <- sum(
    tab$n[!is.na(tab[[category_col]])]
  )

  tab |>
    dplyr::mutate(
      percent = 100 * n / total_n,
      valid_percent = dplyr::if_else(
        valid_n > 0 & !is.na(.data[[category_col]]),
        100 * n / valid_n,
```

```

        NA_real_
    )
}
get_descriptive_stats_over_entire_period(analysis_data,PREGNANT)

```

PREGNANT	n	percent	valid_percent
1	11432	1.00000175	3.0394636
2	362558	31.71436619	96.3944922
7	947	0.08283779	0.2517820
9	1182	0.10339416	0.3142622
NA	767079	67.09940010	NA

```

get_descriptive_stats_over_entire_period(analysis_data,MENTHLTH)

```

MENTHLTH	n	percent	valid_percent
1	39555	3.460030546	3.46006384
2	66084	5.780625928	5.78068155
3	41539	3.633578785	3.63361375
4	20903	1.828467160	1.82848475
5	55130	4.822436708	4.82248311
6	5731	0.501312983	0.50131781
7	21584	1.888036893	1.88805506
8	4351	0.380598986	0.38060265
9	731	0.063943429	0.06394404
10	42260	3.696647475	3.69668304
11	300	0.026242173	0.02624243
12	3036	0.265570794	0.26557335
13	421	0.036826516	0.03682687
14	8309	0.726820726	0.72682772
15	39977	3.496944536	3.49697818
16	697	0.060969316	0.06096990
17	601	0.052571820	0.05257233
18	785	0.068667020	0.06866768
19	128	0.011196661	0.01119677
20	25415	2.223149446	2.22317084
21	1411	0.123425688	0.12342688
22	400	0.034989564	0.03498990
23	271	0.023705430	0.02370566
24	259	0.022655743	0.02265596

25	8695	0.760585655	0.76059297
26	269	0.023530482	0.02353071
27	593	0.051872029	0.05187253
28	2362	0.206613378	0.20661537
29	1140	0.099720258	0.09972122
30	75139	6.572702192	6.57276544
77	15937	1.394071718	1.39408513
88	653219	57.139620608	57.14017042
99	5955	0.520907139	0.52091215
NA	11	0.000962213	NA

```
get_descriptive_stats_over_entire_period(analysis_data,EMPLOY)
```

EMPLOY	n	percent	valid_percent
1	427657	37.4088303	37.6710651
2	72200	6.3156164	6.3598887
3	22839	1.9978167	2.0118213
4	22202	1.9420958	1.9557098
5	90549	7.9206752	7.9761989
6	27679	2.4211904	2.4381629
7	377824	33.0497429	33.2814207
8	83087	7.2679448	7.3188929
9	11203	0.9799702	0.9868398
NA	7958	0.6961174	NA

```
get_descriptive_stats_over_entire_period(analysis_data,INCOME)
```

INCOME	n	percent	valid_percent
1	77835	6.808532	6.808532
2	125785	11.002906	11.002906
3	108922	9.527833	9.527833
4	126922	11.102364	11.102364
5	329376	28.811807	28.811807
6	100829	8.819907	8.819907
7	31201	2.729273	2.729273
9	242328	21.197378	21.197378

```
get_descriptive_stats_over_entire_period(analysis_data,EDUCA)
```

EDUCA	n	percent	valid_percent
1	70463	6.163674184	6.1638144
2	285253	24.952195508	24.9527630
3	324998	28.428846097	28.4294927
4	457453	40.015202966	40.0161131
9	5005	0.437806924	0.4378169
NA	26	0.002274322	NA

```
get_descriptive_stats_over_entire_period(analysis_data, INSURANCE)
```

INSURANCE	n	percent	valid_percent
1	605179	52.937374	52.937374
2	58134	5.085208	5.085208
9	479885	41.977418	41.977418

Around 66% of women report missing responses when asked if they are currently pregnant.
32% of women report they aren't currently pregnant.

Trends in the U.S. Regions Over Time

```
regional_summary <- analysis_data_v2 |>
  select(
    STATE,
    REGION,
    YEAR,
    all_of(variables_to_summarize)
  ) |>
  pivot_longer(
    cols = all_of(variables_to_summarize),
    names_to = "VARIABLE",
    values_to = "VALUE"
  ) |>
  mutate(
    VALUE_CODE = if_else(
      is.na(VALUE),
      "Missing",
      as.character(VALUE)
    ),
```

```

VALUE_LABEL = map2_chr(
  VARIABLE,
  VALUE_CODE,
  function(variable, code) {
    if (code == "Missing") {
      return("Missing")
    }

    label <- unname(variable_labels[[variable]][code])

    if (is.na(label)) {
      paste("Unlabeled code:", code)
    } else {
      label
    }
  }
) |>
count(
  STATE,
  REGION,
  YEAR,
  VARIABLE,
  VALUE_CODE,
  VALUE_LABEL,
  name = "N"
) |>
group_by(STATE, REGION, YEAR, VARIABLE) |>
mutate(
  PERCENT = round(100 * N / sum(N), 2)
) |>
ungroup()

regional_summary

```

A tibble: 19,783 x 8

	STATE	REGION	YEAR	VARIABLE	VALUE_CODE	VALUE_LABEL	N	PERCENT
	<chr>	<chr>	<int>	<chr>	<chr>	<chr>	<int>	<dbl>
1	Alabama	South	2017	EDUCA	1	No high school	463	11.5
2	Alabama	South	2017	EDUCA	2	High school	1195	29.6
3	Alabama	South	2017	EDUCA	3	Some college	1201	29.8
4	Alabama	South	2017	EDUCA	4	College graduate	1168	28.9

5	Alabama	South	2017	EDUCA	9	Don't know	10	0.25
6	Alabama	South	2017	EMPLOY	1	Employed for wages	1279	31.7
7	Alabama	South	2017	EMPLOY	2	Self-employed	151	3.74
8	Alabama	South	2017	EMPLOY	3	Out of work for 1 yea~	89	2.2
9	Alabama	South	2017	EMPLOY	4	Out of work for less ~	79	1.96
10	Alabama	South	2017	EMPLOY	5	Homemaker	378	9.36

i 19,773 more rows

```
plot_variable <- function(data, variable_name, plot_title, legend_title) {

  data |>
    dplyr::filter(VARIABLE == variable_name) |>
    ggplot(
      aes(
        x = factor(YEAR),
        y = PERCENT,
        fill = VALUE_LABEL
      )
    ) +
    geom_col(position = "dodge") +
    facet_wrap(~ REGION) +
    labs(
      title = plot_title,
      x = "Year",
      y = "Percent",
      fill = legend_title
    ) +
    theme_minimal() +
    theme(
      legend.position = "bottom",
      axis.text.x = element_text(angle = 45, hjust = 1)
    )
}

var_by_region_plots <- list(
  marital = plot_variable(
    regional_summary,
    "MARITAL",
    "Marital Status by Region and Year",
    "Marital status"
  ),

  home = plot_variable(
```

```

    regional_summary,
    "HOME",
    "Homeowner Status by Region and Year",
    "Homeowner status"
),

pregnant = plot_variable(
    regional_summary,
    "PREGNANT",
    "Pregnancy Status by Region and Year",
    "Pregnancy status"
),

employ = plot_variable(
    regional_summary,
    "EMPLOY",
    "Employment Status by Region and Year",
    "Employment status"
),

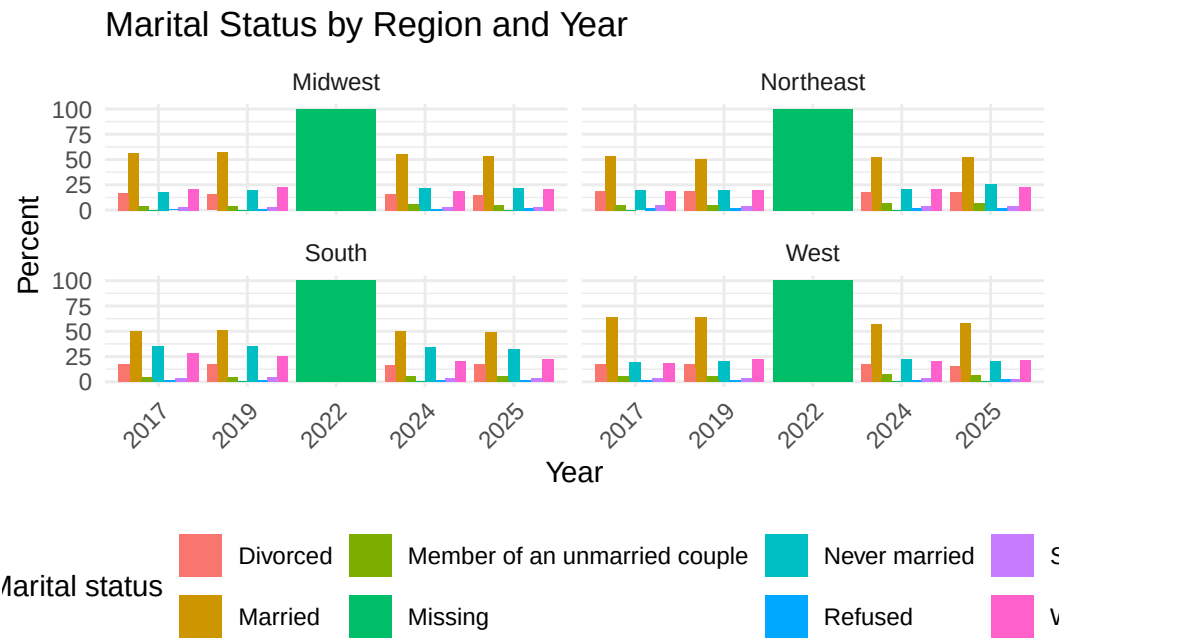
income = plot_variable(
    regional_summary,
    "INCOME",
    "Income Categories by Region and Year",
    "Income category"
),

insurance = plot_variable(
    regional_summary,
    "INSURANCE",
    "Health Insurance Status by Region and Year",
    "Insurance status"
),

race = plot_variable(
    regional_summary,
    "RACE",
    "Race and Ethnicity by Region and Year",
    "Race and ethnicity"
)
)

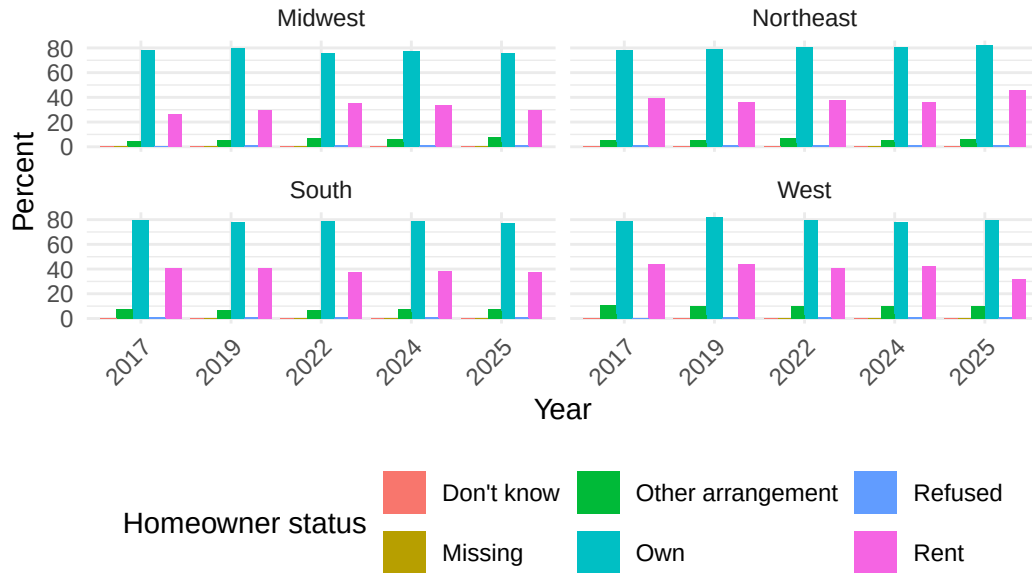
```

```
var_by_region_plots$marital
```



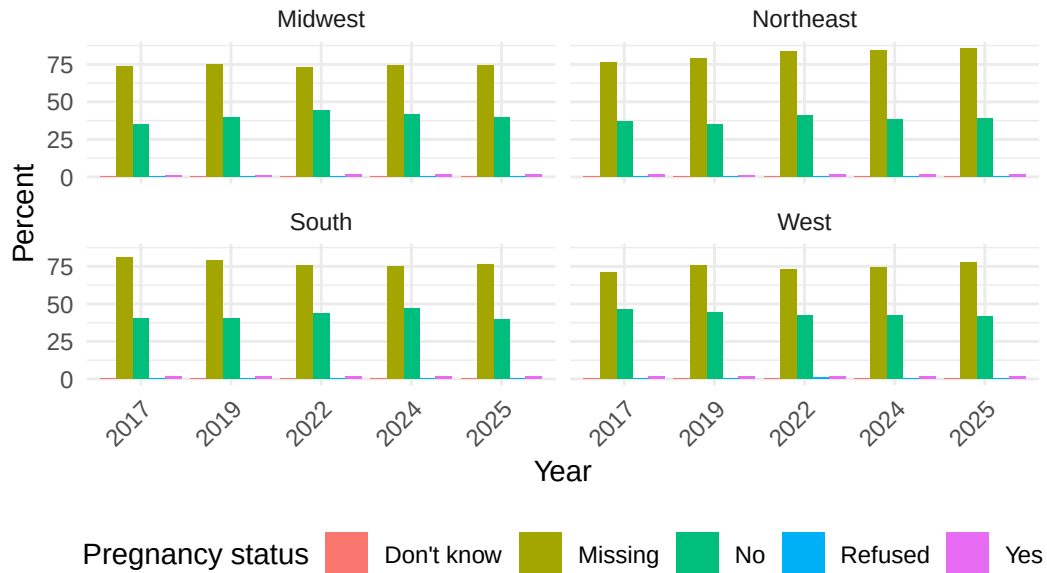
```
var_by_region_plots$home
```

Homeowner Status by Region and Year

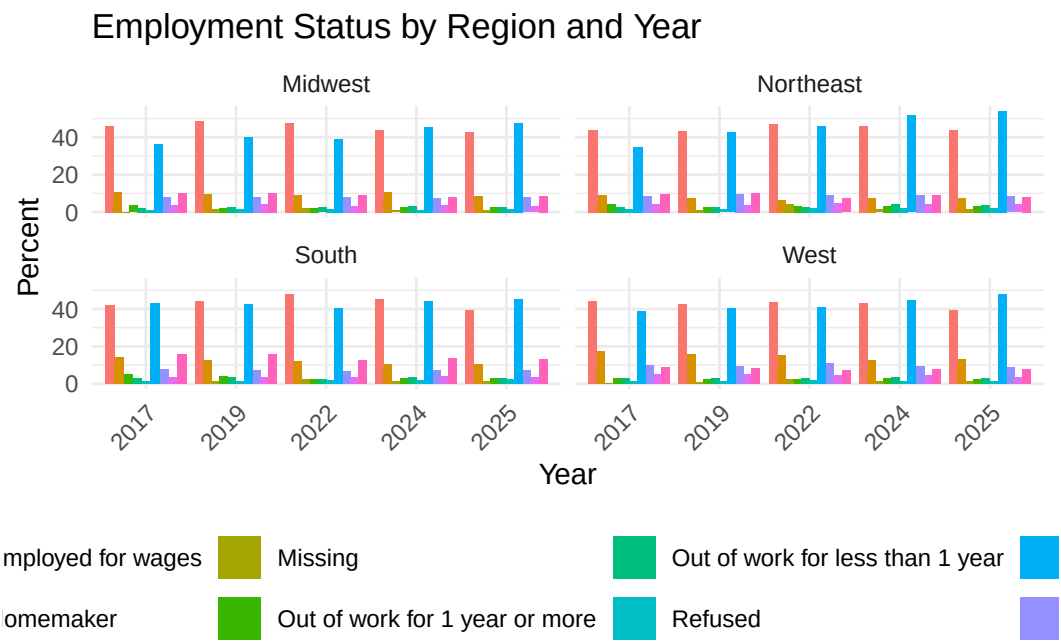


```
var_by_region_plots$pregnant
```

Pregnancy Status by Region and Year

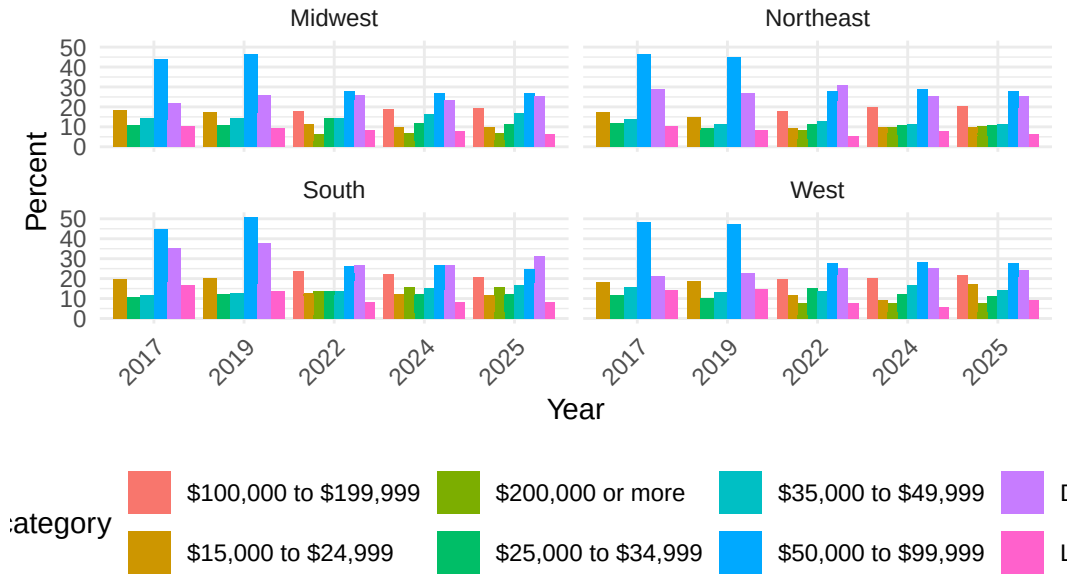


```
var_by_region_plots$employ
```



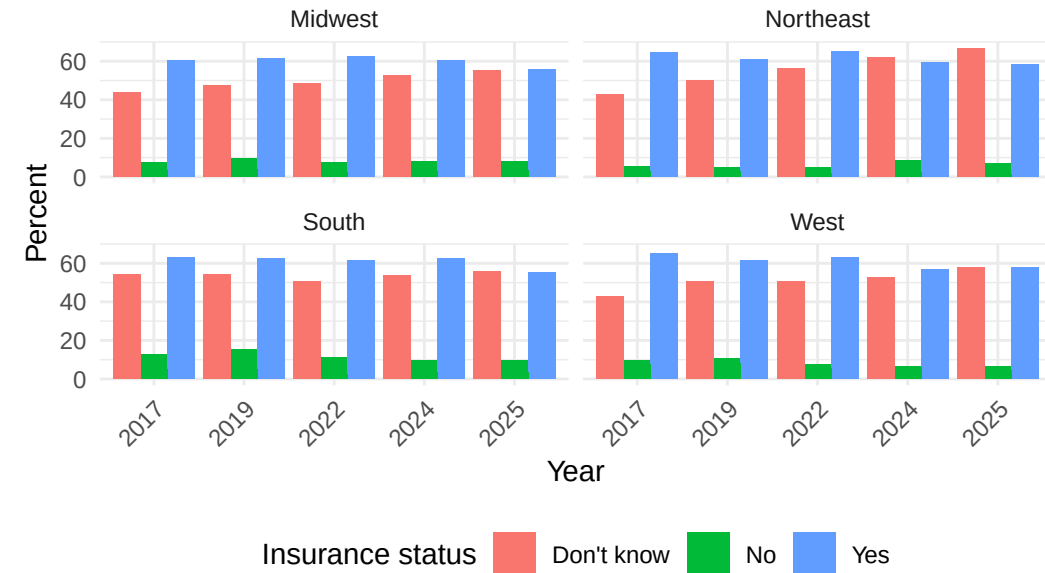
```
var_by_region_plots$income
```

Income Categories by Region and Year



```
var_by_region_plots$insurance
```

Health Insurance Status by Region and Year



```
var_by_region_plots$race
```

